

Decode Labs Internship

Encryption and Decryption Using Caesar Cipher and ZeeCipher

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Task Week:	02
Program:	Cyber Security
Submission Date:	11 th April 2026

DECODE LABS

Encryption and Decryption Using Caesar Cipher

Objective

The objective of this task was to implement encryption and decryption techniques using Python. The program provides a menu-driven interface where users can encrypt and decrypt messages using both a standard Caesar Cipher and a custom cipher developed by me called ZeeCipher.

Features

- Caesar Cipher Encryption
- Caesar Cipher Decryption
- Custom ZeeCipher Encryption
- Custom ZeeCipher Decryption
- Support for uppercase and lowercase letters
- Preserves spaces, numbers, and special characters
- Menu-driven user interface

```
-----  
-----WELCOME TO ZEENCRYPT-----  
-----  
-----  
Enter Your Choice.....  
What You Want To Do?  
1.Ceaser Cypher  
2.ZeeCypher  
3.Exit
```

Caesar Cipher

Caesar Cipher is one of the simplest and oldest encryption techniques. It works by shifting each alphabetic character in a message by a fixed number of positions. The resulting text becomes unreadable without knowing the shift value used during encryption.

Caesar Cipher Encryption

The encryption function takes a message and a user-defined shift value. Each letter is shifted forward by the specified number of positions in the alphabet while preserving spaces, numbers, and special characters.

Screen Shot:

```
-----Ceaser Cypher-----  
  
Enter Your Choice :  
1.Encrypt.  
2.Decrypt  
3.Exit  
1  
Enter the message below which you want to encrypt(symbols and numbers will not be encrypted.....):  
zee  
Enter the shift you want to give to your text...(must be an integer)1  
  
Your Encrypted Text Is: aff
```

Caesar Cipher Decryption

The decryption function reverses the encryption process by shifting characters backward using the same shift value that was used during encryption.

```
-----Ceaser Cypher-----  
  
Enter Your Choice :  
1.Encrypt.  
2.Decrypt  
3.Exit  
2  
Enter the message you want to decrypt :  
aff  
Enter the shift you gave while encrypting this text.....:1  
  
Your Decrypted Text Is: zee
```

My Custom Algorithm: ZeeCipher

In addition to the standard Caesar Cipher, I designed my own encryption algorithm called ZeeCipher.

Instead of requiring the user to provide a shift value, ZeeCipher automatically generates the shift using the formula:

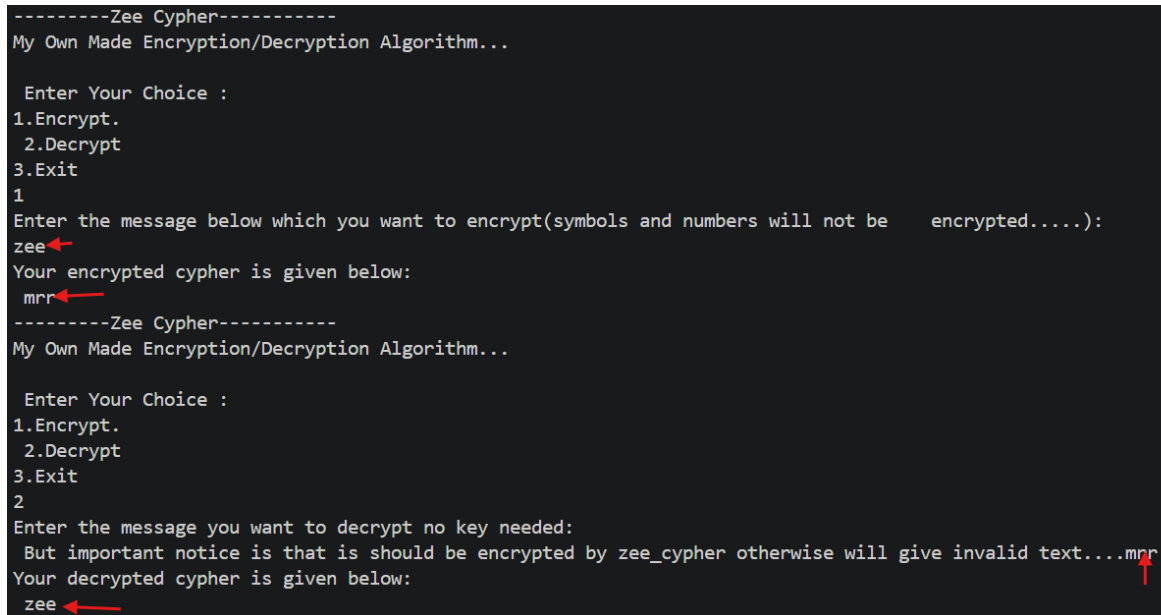
$$\text{Shift} = (\text{Length of Message} + 10) \% 26$$

This makes the encryption process automatic and more user-friendly while demonstrating custom algorithm design and implementation.

Example:

```
Original Text: zee  
Length = 3  
Shift = (3 + 10) % 26 = 13  
Encrypted Text: mrr  
Decrypted Text: zee
```

Output Screenshots



```
-----Zee Cypher-----  
My Own Made Encryption/Decryption Algorithm...  
  
Enter Your Choice :  
1.Encrypt.  
2.Decrypt  
3.Exit  
1  
Enter the message below which you want to encrypt(symbols and numbers will not be encrypted....):  
zee  
Your encrypted cypher is given below:  
mrr  
  
-----Zee Cypher-----  
My Own Made Encryption/Decryption Algorithm...  
  
Enter Your Choice :  
1.Encrypt.  
2.Decrypt  
3.Exit  
2  
Enter the message you want to decrypt no key needed:  
But important notice is that is should be encrypted by zee_cypher otherwise will give invalid text...mrr  
Your decrypted cypher is given below:  
zee
```

Learning Outcomes

- Understanding of basic cryptography concepts
- Implementation of Caesar Cipher in Python
- Working with ASCII values using ord() and chr()
- Use of functions and loops
- Design and implementation of a custom encryption algorithm

Conclusion

This task successfully demonstrates encryption and decryption using both the traditional Caesar Cipher and my custom-developed ZeeCipher. The task enhanced my understanding of cryptographic techniques, Python programming, and algorithm design.